PERFORM BLOOD CULTURE COLLECTIONS

Introduction

In this lesson you will learn how to perform a blood culture collection. A blood culture test is collected to determine if a patient has a bacterial infection. It also aids in the diagnosis of a bacterial infection by showing the type of bacteria that grew in the cultures. The steps for a blood culture are similar to a routine venipuncture, differing by the supplies and the type of skin preparation that must occur. You will learn how to select the proper supplies and confirm their quality. You will also learn the blood culture collection steps and guidelines for performing this procedure. This procedure, like many others, requires you to observe the patient before, during, and after the collection.

Blood Cultures

To help determine if a patient has an infection, a phlebotomist collects blood cultures to confirm the presence of bacteria. A provider may order this test when a patient has a fever of unknown origin. To perform this test, blood is collected into bottles that contain a ***broth*** that encourages bacteria to grow. These bottles are then transported to the laboratory. There, the laboratory personnel monitor the sample for typically 2 to 3 days. In this waiting time, if the patient has a bacterial infection, it will be detectable when the laboratory processes the sample. If the sample has been contaminated by bacteria from another source, the laboratory will not be aware of this and will assume that the bacteria was from the patient who had been tested.

broth. An additive that is rich in nutrients and helps encourage the growth of bacteria in the blood.

Blood Culture Preparation

Blood cultures are collected to check for bacteria levels in the patient, so it is important to follow all of the precautions to avoid cross contamination from you, the surroundings, or the patient. Review the requisition to determine if one or two collections will be necessary. Depending on what the ordering provider has requested, you may be required to collect two sets of blood cultures at separate times.

To begin performing a blood culture test, review the requisition to guide you in the collection of the necessary supplies. Blood cultures are collected into bottles that contain vacuum and additives, just like a routine venipuncture tube, but they differ by their size and the volume of additive. Blood cultures consist of two bottles: one ***aerobic*** and one ***anerobic***. These bottles contain a broth that provides a medium for the bacteria cultures to grow in. This additive, like the ones in evacuated tubes, must be inverted to help ensure that it is properly mixed with the blood.

aerobic. With oxygen.

anerobic. Without oxygen.

Performing the Blood Culture Collection

When gathering supplies, confirm that they are not expired and that any sealed package is not damaged or ripped or have any signs of use. Like in venipuncture, supplies include gauze, alcohol wipes, a sharps container, and the necessary PPE, but there are special supplies unique to the collection of blood cultures.

Supplies:

- Two blood culture bottles: one aerobic and one anerobic
- At least four alcohol wipes
- Other antimicrobial cleanser
- Gauze
- Butterfly needle
- Adapter
- Tourniquet
- Sharps container
- Other evacuated tubes required for other tests
- Required PPE

Blood Culture Preparation Steps

Keep the collected blood as free from contamination as possible. One of the methods of achieving this is to perform blood cultures first before any other vacuum tube. If it is necessary to collect other tubes for other tests, they can be collected after the blood cultures are performed.

Blood cultures can be collected from the antecubital region or the dorsal part of the hand. Choose a vein that is large enough and that will be able to withstand the blood removal requirements of performing this test.

Blood Culture Steps

1. Review the requisition to determine if one or two blood cultures will be required.
2. Wash your hands and don gloves.
3. Determine the required amount of blood from the requisition. Typically, 20 to 30 milliliters (mL) of blood is collected from adults. This is divided equally between the two bottles.
4. With a smear-free marker, mark the level of blood that should be collected in each bottle. The mark is made either 10 or 15 mL above the level of the additive.
5. Choose an adapter that will fit on the top of the blood culture bottles. Check the packaging for an expiration date or rips or tears before use.
6. Review what antiseptics are required for the test and their manufacturer's instructions. Confirm that you know how to effectively use them, even if you have used them before.
7. Gather any other evacuated tubes needed for other ordered tests
8. Determine the proper order of draw.

After properly introducing yourself and identifying the patient, the blood culture collections can begin.

Calming Patient Concerns

Patients may express concern about having blood cultures collected. Since the bottles are larger than the traditional venipuncture tubes, it may appear that a large amount of blood is being collected. Help reduce their fears by explaining that you will be collecting approximately 2 tbsp of blood for the blood cultures instead of saying 30 mL. Some patients may not be familiar with the metric system, so explaining the amount of blood in a way that is easier to translate may help allay any fears.

A quick calculation to remember is that 5 mL of blood equals 1 tsp. Therefore, when the blood cultures require 30 mL of blood, that would be equal to 6 tsp. Converting the amount of blood to teaspoons can be demonstrated to your patient to help them understand how much blood will be collected.

Setup

1. Wash hands and don PPE.
2. Explain to the patient that this test will be slightly different than other venipunctures that they have experienced before. Ask your patient if they have had any negative reaction from the antiseptic that will be used to perform the collection. Allow the patient time to answer any questions regarding this test.
3. Set up supplies within easy reach.
4. Prepare each blood culture bottle by removing just the cap, taking care not to remove the top completely. Wipe each stopper with one alcohol wipe. Allow the alcohol to air dry.

Prepare the Site

1. Apply the tourniquet.
2. Palpate to find an assessable vein.
    - The Clinical and Laboratory Standards Institute (CLSI) recommends an upper extremity venipuncture site due to increased risk of patient injury and culture contamination.
3. Cleanse the chosen site with friction for 30 seconds using a sterile alcohol pad, and allow the site to air dry.
4. Apply the second disinfectant.
5. Do not touch the chosen site, even with gloved hands, after it has been cleansed.

The site must be cleansed differently from the methods used for venipuncture. Traditionally, a scrub of a minimum of 30 seconds with an antiseptic will help remove a majority of the bacteria in the area. Depending on your facility, different antiseptics may be used, including povidone iodine, tincture of iodine, chlorhexidine gluconate, or a combination of povidone iodine and 70% ethyl alcohol.

Preparing the Blood Culture Site

Use the antiseptics and instructions recommended by the manufacturer to help reduce the risk of cross contamination. Skipping steps may increase the risk of not removing bacteria from the site and contaminating the samples.

Perform the Puncture

1. Uncap the butterfly needle.
2. Insert into the chosen vein with a 15° to 30° angle.
3. When flash is seen, attach the aerobic bottle and fill it to the marked level.
4. After it is filled, remove the bottle from the needle.
5. Invert to mix the additive.
6. Release the tourniquet before 1 minute has passed.
7. Insert the second anerobic bottle.
8. Fill to the required level.
9. Continue with other required evacuated tubes or remove the needle and follow with the routine venipuncture post care.
10. Record the exact time that the blood cultures were collected.
11. Perform post venipuncture procedures, including bandaging.
12. Thank the patient.
13. Calculate the time for the next collection (when two collections are required).

Once the blood culture bottles have been filled and inverted, they must be labeled as well. Follow the labeling requirements for venipuncture. Transport to the laboratory or prepare for shipment.

If it is recommended that another blood culture test is to be collected, perform it similarly to the first collection at the exact time that the second collection is ordered. When possible, choose a vein in the other arm or from a different site on the same arm.

If a patient's veins are fragile, it may be necessary to use a syringe to collect their blood. Use this technique only when your facility permits the usage of syringes. When using a syringe, fill the aerobic bottle first. Confirm that all syringe supplies are in their original package and not opened or used. Do not use any expired supplies when performing a blood culture collection. When less than the recommended blood volume is drawn for culture, the blood should be injected into the aerobic bottle first. Any remaining blood should be inoculated into the anaerobic bottle. Injecting blood culture bottles in this manner is important.

Assisting the Patient

Povidone iodine can be irritating to some patients. After the blood cultures have been completed, wipe the excess iodine off their skin before bandaging. Doing this will help reduce the chance of skin irritation.

Conclusion

Blood cultures differ from routine venipuncture not only in the supplies that are used, but also in the cleansing of the patient and handling of the supplies. Take the time to explain the required steps of this procedure and answer any questions that they may have. The proper use of antiseptics helps ensure that only any bacteria existing in the patient's blood appears in the sample. Observing the patient's condition throughout the procedure and afterwards helps ensure their safety. Wiping off any excess antiseptic helps reduce the chances of skin irritation. Following all the required steps in every blood culture test performed helps provide accurate test results.

5.1 CHALLENGE

1. Place the following options in the proper order of draw with a butterfly.
   A. Anaerobic blood cultures bottle
   B. Aerobic blood cultures bottle

5.1 CHALLENGE KEY

1. 1 = B. 2 = A. Blood cultures must be collected before other tests to help reduce the risk of cross contamination of the sample. The aerobic bottle is collected first because it is acceptable to have air in the sample and the butterfly needle has air in the tubing.